

SHETH L.U.J AND SIR M.V COLLEGE

Artificial Intelligence

Practical No.4

AIM: Feed Forward Backpropagation Neural Network

- Implement the Feed Forward Backpropagation algorithm to train a neural network.
- Use a given dataset to train the neural network for a specific task.
- Evaluate the performance of the trained network on test data.

Code:

```
pre7.py - C:\Users\gazala\OneDrive\Desktop\pre7.py (3/3/2)
File Edit Format Run Options Window Help

import matplotlib.pyplot as plt
import pandas as pd

from sklearn.metrics import accuracy_score, classification_report
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.preprocessing import StandardScaler

# Load Iris dataset
df = pd.read_csv("Iris.csv")

# Remove extra spaces from column names
df.columns = df.columns.str.strip()

# Separate input features and target
X = df.drop(["Species", "Id"], axis=1)
y = df["Species"]

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Standardize the input features
scaler = StandardScaler()
```

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```
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# Feed Forward Backpropagation Neural Network
model = MLPClassifier(
    hidden_layer_sizes=(16, 8),
    activation="relu",
    solver="adam",
    max_iter=300,
    random_state=42
)

# Train the neural network
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Evaluate the model
print("Accuracy:", accuracy_score(y_test, y_pred) * 100, "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Plot the loss curve
plt.plot(model.loss_curve_)
plt.title("Feed Forward Backpropagation Loss Curve")
plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.show()
```

Output:

```
IDLE Shell 3.13.2
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
----- RESTART: C:/Users/gazala/OneDrive/Desktop/pno.7.py -----

Warning (from warnings module):
  File "C:\Users\gazala\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\network\_multilayer_perceptron.py", line 785
    warnings.warn(
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (300) reached and the optimization hasn't converged yet.
Accuracy: 93.33333333333333 %

Classification Report:

```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	0.90	0.90	0.90	10
Iris-virginica	0.90	0.90	0.90	10
accuracy			0.93	30
macro avg	0.93	0.93	0.93	30
weighted avg	0.93	0.93	0.93	30

```
>>>
```

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